

Sin

Sin is a traditional first person shooter centred around rescuing a group of hostages. Installation-wise, this game was one of the simplest—just run the `'SineDemo_x86.run'` script at the command prompt. This creates a directory called 'SinDemo' in the current directory. Change to this directory, and then start the command by typing `'Sin'` at the command prompt.

America's Army

America's Army is developed by the US army, and is based on the Unreal engine. It has a graphical installation utility—just download `armyops-lnx-1702.sh.bin` from www.americasarmy.com; this is the installer script that invokes the installer GUI. To run the script, open a terminal window with `'xterm'`, `'cd'` to the directory containing the script and type `'./armyops-lnx-1702.sh.bin'` at the command prompt. If



The graphical install utility for America's Army is similar to installing games in Windows

the downloaded file does not have executable rights, you will have to manually set it using the command `'chmod a+x armyops-lnx-1702.sh.bin'`.

Asteroids3D

Asteroids3D is the OpenGL implementation of the popular 2D asteroids game you find on every Linux system. Just download the compressed tarball from the Web site, and compile it using the `'make'` command. The prerequisite for this game is that OpenGL should be installed on your system. Moreover, it must be installed at the default location (`/usr/local/openssl`), if you want `'make'` to work. Otherwise, you'll have to edit the Makefile that comes along with the source code. Open the Makefile in a text editor and edit the lines starting with `'INC'` and `'LIB'`—the `'INC'` stands for `'Include'`, and specifies the location of the OpenGL files to be included; the `'LIB'` stands for `'Libraries'`, and specifies the location of the OpenGL libraries. Modify the `'INC'` section from `'INC=-I/usr/X11R6/include -I/usr/local/openssl/include'`

to `'INC=-I/usr/X11R6/include -I/your_opengl_directory_path/include'`
Modify the `'LIB'` section from `'LIB=-lglut -lGLU -lGL -lm -lX11 -lXext -lXmu -lXi -lpthread -L/usr/X11R6/lib -L/usr/local/openssl/lib'`
to `'LIB=-lglut -lGLU -lGL -lm -lX11 -lXext -lXmu -lXi -lpthread -L/usr/X11R6/lib -L/your_opengl_directory_path/lib'`

The handwriting on the wall

Gaming on Linux is currently not the hottest trend, and the available games are primitive and unsophisticated

Old warhorses

Some games have been available on Linux for ages, and still are. These are the *Solitaires* of the Linux world, but obviously with the Linux twist—they are more difficult than their Windows counterparts. These include games such as *Sokoban*, *Xboing*, *Asteroids*, etc.

Xboing is a breakout game that is quite popular. It's got lots of extra features, such as bullets that not only blow up the bricks, but also the ball used to break the bricks. Then there's the occasional skeleton that eats up the balls, and the paddle shrinking powers, as well as some troublesome features such as disappearing walls—you can't bounce off the walls, and end up running all over the place to get a few points.

Sokoban is a real brain teaser. In this game, you have to move diamonds around from their initial position to the



Play your favourite classic games in Linux

final highlighted spots. Easier said than done, though—even the basic levels can be taxing at times.

compared to their Windows counterparts. Installing and configuring them is also more complicated. But all said and done, gaming on Linux definitely seems to be going along the right direction. It's only a matter of time before all this catches up and reaches the level currently available on Windows. All the libraries are in place and driver support from vendors is also being improved. The only thing remaining is game developers getting serious about Linux, and considering it a viable platform to leverage their skills. That, too, seems to be getting better by the day. Commercial gaming companies might take a while to warm up to Linux, since numbers are really important to them, and selling games for Linux will be profitable only if there are enough Linux users. With the increasing refinement of Linux, and its consequent adoption, gaming on Linux should be something to look out for in the near future. ■

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Vroom, Vroom, Whoosh, Zing, Crash, Kaboom!

Environmental audio is used by game developers to give those amazing effects that are produced when a car whizzes past you in a car race, or if a rocket passes by in a first person shooter. To generate these effects, programmers use various specialized audio APIs (Application Programming Interfaces), which provide an easy and uniform way of interfacing with the sound card. For example, these APIs provide simple functions to create the effects of attenuation of sound with distance, or implementing the Doppler effect, based on the relative velocities of the source and the tar-

get. Some of the popular APIs available on Windows are Direct Audio (part of the DirectX suite), EAX, A3D and OpenAL. Of these, OpenAL has been successfully ported to Linux, and is currently used by many games to achieve environmental audio effects. Currently, OpenAL is available for download from Creative but not via HTTP or FTP; you have to login to a CVS repository (the place where they keep all the source code), and download the source code. Most of you will be saved the hassle, though, as almost all Linux distributions come bundled with OpenAL.